

# Wenlong Ding (Curriculum Vitae)

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## EDUCATION

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### The Chinese University of Hong Kong (CUHK)

Shatin, Hong Kong SAR, China

*Ph.D. in Computer Science and Engineering*

08/2021 – Present

- Thesis (Ongoing): Learning-Inspired Network Traffic and Configuration Management.

### Huazhong University of Science and Technology (HUST)

Wuhan, Hubei, China

*B.E. in Computer Science and Technology (ACM class)*

09/2017 – 06/2021

- GPA: 90.64/100 or 3.96/4.0 | Rank: Top 5%

## RESEARCH INTERESTS

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### Topic 1: Learning/LLM-Driven Network Configuration Management

05/2023 – Present

- **Agentic Routing Path Analysis (RPA) at Scale:** Benchmark [C1]; agentic RPA system for global production datacenter networks [A2\*].
- **Intent-Based Configuration:** ACL configuration generation and deployment [A1]; configuration template database [C3].

### Topic 2: Learning-Based Traffic Engineering (TE) Systems

08/2021 – 04/2023

- Link failure restoration in TE [C2]; TE acceleration [J1].

## PUBLICATION

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### Conference Papers

[C1] **Wenlong Ding**, Zhixiong Niu, Jianan Yang, Fajun Zhang, Bo Zhang, Ling Liang, Yongqiang Xiong, Tianyin Xu, and Hong Xu. RoutingBench: Can Agentic Routing Analysis Scale to Production Datacenter Networks? In *Advances in Neural Information Processing Systems (NeurIPS), Evaluations and Datasets Track*, 2026 (To appear).

[C2] **Wenlong Ding** and Hong Xu. Dynamic Learning-based Link Restoration in Traffic Engineering with Archie. In *Proceedings of the IEEE International Conference on Computer Communications (IEEE INFOCOM)*, 2024.

[C3] **Wenlong Ding**, Libin Liu, Li Chen, and Hong Xu. ABC: Automatic Bottom-up Construction of Configuration Knowledge Base for Multi-Vendor Networks. In *Proceedings of the Fifth IEEE International Conference on Cognitive Machine Intelligence (IEEE CogMI) (Invited Paper)*, 2023.

[C4] Jianqiang Li, **Wenlong Ding**, Zhixiong Niu, Yongqiang Xiong, and Hong Xu. Saving Lost Intents in Network Configurations. In *Proceedings of the 10th Asia-Pacific Workshop on Networking (ACM APNet)*, 2026.

### Journal Papers

[J1] **Wenlong Ding**, Libin Liu, Li Chen, and Hong Xu. Learning for Accelerated Traffic Engineering with Differentiable Network Modeling. In *IEEE Transactions on Network Science and Engineering (TNSE)*, vol. 13, pp. 25–36, 2026.

### Posters

[P1] **Wenlong Ding**, Jianqiang Li, Zhixiong Niu, Huangxun Chen, and Hong Xu. Poster: Automating Network Configuration with Natural Language Intents. In *Proceedings of the ACM SIGCOMM 2024 Conference: Posters and Demos (ACM SIGCOMM Posters and Demos)*, 2024.

## Preprints and Manuscripts Under Review

[A1] (Under review) **Wenlong Ding**, Jianqiang Li, Zhixiong Niu, Huangxun Chen, Yongqiang Xiong, and Hong Xu. Automating Conflict-Aware ACL Configurations with Natural Language Intents. *arXiv preprint arXiv:2508.17990*, 2025.

[A2\*] (Under review) TARS: Scaling Agentic Routing Analysis for Production Datacenter Networks.

- Scaling agent skills for *RPA* across global production datacenter networks through context management and symbolic reflection, guided by operator usability requirements and operational experience.

[A3\*] (Under review; *non-first author*) Adaptive Route Scaling for Cloud WANs.

- Reducing FIB entries while maintaining effective load balancing across the global cloud WAN.

\* *These projects were conducted by **Wenlong Ding** during his internship at Microsoft Research Asia (MSRA), in collaboration with Azure network operators.*

## INTERNSHIP

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**Research Intern — Microsoft Research Asia, Beijing, China**

*02/2025 – 07/2026*

- *Lead role*: Resolved RPA scalability challenges in global datacenter networks to support Azure network operators' daily analysis and diagnosis [C1, A2\*].
- *Contributing role*: Assisted with writing a paper for the project on FIB optimization to reduce memory pressure in Azure's global WAN. [A3\*]

## SERVICE AND TEACHING

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### Academic Services

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|--------------------------------------|-----------------------------------------------------------------------------|
| <b>Reviewer</b>                      | IEEE Transactions on Network Science and Engineering.                       |
| <b>Shadow TPC</b>                    | ACM CoNEXT 2025 2026.                                                       |
| <b>Artifact Evaluation Committee</b> | USENIX NSDI 2026 2027, ACM SIGCOMM 2026, ACM EuroSys 2026, ACM ASPLOS 2026. |

### Teaching Assistant at CUHK

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|----------|---------------------------------------------------------------------------|
| CSCI4430 | <i>Data Communication and Computer Networks</i> . Spring 2023; Fall 2023. |
| ENGG2020 | <i>Digital Logic and Systems</i> . Fall 2022; Spring 2024.                |
| CSCI3170 | <i>Introduction to Database System</i> . Fall 2024.                       |
| ENGG1103 | <i>Multivariable Calculus for Engineers</i> . Spring 2022.                |
| ENGG2760 | <i>Probability for Engineers</i> . Fall 2021.                             |

## HONORS AND AWARDS

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|----------------|---------------------------------------------------------------------------------------------|
| <b>06/2021</b> | Excellent Undergraduate Thesis (3%), HUST                                                   |
| <b>05/2021</b> | Outstanding Graduate and Honours Bachelor's Degree (3%), HUST                               |
| <b>09/2020</b> | Merit Student (5%), HUST                                                                    |
| <b>12/2019</b> | Outstanding Individual in University Students Innovation and Entrepreneurship Project, HUST |
| <b>04/2019</b> | Meritorious Winner, Mathematical Contest in Modeling (5%), COMAP, USA                       |